

$$v_f = \sqrt{v^2 - eEx/u}$$

**21 D**

$$R_{eff} = R_J + R_K = \frac{\rho 2x}{\frac{x}{2}t} + \frac{\rho x}{xt}, \text{ where } \rho \text{ is resistivity and } t \text{ is thickness}$$

$$R_{eff} = R + 4R = 8.0 + 4(8.0) = 40 \, \Omega$$

Option A: Student thought that J and K are connected in parallel.

Option B: Student thought that the resistance of K is also  $8.0 \, \Omega$

Option C: Student forgot to add the resistance of J.